Assignment 2: Data Cleaning and Transformation

Dataset Overview & Missing Value Treatment Strategy

ASSIGNMENT QUESTIONS

- **Handling Missing Values (Price):** Check for missing values in the 'Price' column. How would you handle products with missing price information?
- **Handling Missing Values (Category):** If there are products with missing categories, propose a strategy to impute or deal with these missing values effectively.

1. Complete Dataset (Rows 1–34)

#	Product ID	Product Name	Brand Name	Price	Quantity	Category (F)	Price range Imputed (G)	Category Imputed (H)
1	28-JAN-US	laptop	Dell	1000	30	Electronics	1000	Electronics
2	15-FEB-US	Sneakers	Nike	80	15	Fashion	80	Fashion
3	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	130	Kitchen
4	11-APR-US	smartphone	Samsung	900	25	Electronics	900	Electronics
5	22-MAY-US	Backpack	North Face	70	20	*[Blank]*	70	Electronics
6	07-JUN-UK	Headphones	Sony	*[Blank]*	45	Electroni	309	Electroni
7	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	30	Fashion
8	23-AUG-UK	Blender	Ninja	90	35	Kitchen	90	Kitchen
9	05-SEP-UK	Tablet	Apple	500	50	Electronics	500	Electronics
10	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	130	Outdoor
11	17-JUN-IN	Laptop	HP	950	25	Electronics	950	Electronics
12	25-NOV-AU	Sneakers	Adidas	90	40	*[Blank]*	90	Electronics
13	08-DEC-DE	Coffee Maker	Nespresso	120	35	*[Blank]*	120	Electronics
14	18-FEB-CA	Smartwatch	Fitbit	150	15	Electroni	150	Electroni
15	16-APR-ES	headphones	Bose	250	20	Electroni	250	Electroni
16	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	50	Accessories
17	20-AUG-CN	Smartwatch	Huawei	160	15	Electroni	160	Electroni
18	27-JAN-IT	laptop	Asus	980	10	Electronics	980	Electronics
19	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	150	Fashion
20	14-AUG-US	Camping Tent	Coleman	*[Blank]*	10	Outdoor	309	Outdoor
21	14-MAY-RU	Camera	Nikon	700	50	Electronics	700	Electronics
22	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	80	Kitchen
23	19-JUL-BR	Fitness Tracker	Xiaomi	150	30	*[Blank]*	150	Electronics
24	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	50	Accessories
25	29-SEP-CA	Smartphone	Google	800	45	Electronics	800	Electronics
26	03-JUN-CA	Sunglasses	Ray-Ban	*[Blank]*	25	Fashion	309	Fashion
27	11-JUL-CA	Blender	Vitamix	400	40	Kitchen	400	Kitchen
28	16-APR-ES	headphones	Bose	250	20	Electroni	250	Electroni
29	07-MAR-CA	Dress	Zara	60	30	Fashion	60	Fashion
30	13-APR-CA	Toaster	Hamilton	40	10	Kitchen	40	Kitchen
31	24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics	130	Electronics
32	02-DEC-CA	Jeans	Levi's	50	50	Fashion	50	Fashion
33	17-JUN-IN	Laptop	HP	950	25	Electronics	950	Electronics
34	09-JUL-FR	Watch	Casio	100	20	Accessories	100	Accessories

2. Handling Missing Values Analysis & Strategies

A. Question 1: Missing Values in 'Price'

Audit: 3 products have missing prices (Sony Headphones, Coleman Camping Tent, Ray-Ban Sunglasses).

Spreadsheet Formula Used: `=ROUND(IF(ISBLANK([@price]),AVERAGE([price]),[@price]),0)`

Evaluation & Recommendations:

- **Global Average ($309):** Replaces blanks with overall dataset mean, which distorts specific item values (e.g., pricing a tent equal to luxury sunglasses).
- **Category-Based Mean:** Calculate average price within each product category (e.g., mean price of *Fashion* for sunglasses).
- **Historical / Match Lookup:** Match against identical product entries (e.g. Bose headphones at $250).

B. Question 2: Missing Values in 'Category'

Audit: 4 products have missing category values (North Face Backpack, Adidas Sneakers, Nespresso Coffee Maker, Xiaomi Fitness Tracker).

Spreadsheet Formula Used: `=IF(ISBLANK([@Category]),"Electronics",[@Category])`

Evaluation & Recommendations:

Strategy	Description	Example Outcome
Static Fallback (Current)	Fills missing entries with "Electronics"	Mistakenly marks Backpack & Coffee Maker as Electronics
Keyword Mapping	Derive category from `Product Name`	"Coffee Maker" → Kitchen, "Sneakers" → Fashion
Brand Lookup	Map brand name to main product line	North Face → Outdoor, Nespresso → Kitchen
Text Clean-up	Fix spelling inconsistencies	Normalize "Electroni" → "Electronics"